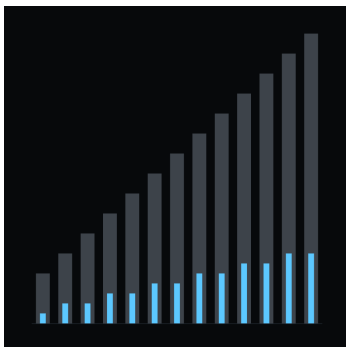




MENGER DIAGONAL SLICES: A RECURRENCE OF ORDER $\lceil D/2 \rceil$

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ABSTRACT. Slice a Menger sponge along its main diagonal and count the cubes the cut meets; then do the same in every dimension. The count obeys a linear recurrence with constant coefficients of order at most $\lceil D/2 \rceil$ in dimension D , a fourfold sharpening of the order $2D + 1$ that the standard carry automaton gives for free; exactness of that order is verified for $2 \leq D \leq 24$. A discrete Fourier extraction then turns a relaxed census into an exact product of cosines, and the formula decides half of the sign law this cut obeys: at every odd dimension D with $D \not\equiv 1 \pmod{3}$, the counting exponent sits strictly above the solid's dimension minus one, proved, with the parity carried by the single factor $(-1)^{D-1}(D-1)$ that every term of the formula ends in. Two exact identities frame what remains: the dominant root is pinned to within $2(D-1)/3$ of $f_D/3$ unconditionally, so the slice exponent converges to the generic Marstrand–Mattila value in every case, and the whole sign law is equivalent to the parity-free statement that the Perron carry vector puts more than a third of its mass on carries divisible by three. At $D = 3$ the machine outputs $\begin{pmatrix} 6 & 6 \\ 1 & 3 \end{pmatrix}$, reproducing the published hexagon-triangle substitution spectrum and the integers 1, 6, 42, 306, 2250, 16578, 122202 by a route that never mentions a hexagon.

1. INTRODUCTION

Take a cube. Cut it into 27 equal subcubes and throw away the seven that touch the middle of a face or the middle of the cube; twenty survive. Repeat inside each survivor forever, and the limit is the Menger sponge. Now slice the sponge with the plane through its centre perpendicular to a main diagonal. The picture that falls out is famous: a hexagon full of six-pointed stars, first rendered by Pérez-Duarte [12], popularised by Hart

[5] and by Cook’s slicing code [3], and given its substitution rule and its dimension 1.8184 by Abel [1].

This paper asks the same question one dimension at a time. In dimension D the sponge becomes the set of cells with at most one coordinate digit equal to the middle digit 1, and the diagonal cut becomes the hyperplane $\sum_i x_i = D(3^L - 1)/2$ through the centre of the level- L grid. Let $a_D(L)$ count the cells the hyperplane meets. The question is how much memory this sequence has: how many previous terms determine the next one.

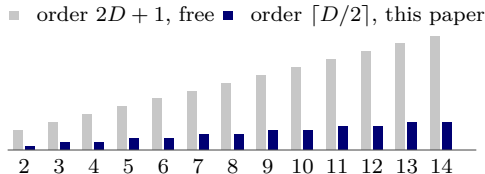


FIGURE 1. How much memory the slice census needs, by dimension D .

The answer is Theorem 3.1: at most $\lceil D/2 \rceil$ terms, in every dimension at once. Figure 1 is that half of the contribution in one picture; the other half is Theorem 3.18, a proved sign law at odd dimension. Three moves prove the order. First, the digit polynomial factors,

$$P_D(t) = (1 + t^2)^{D-1}(1 + Dt + t^2),$$

which is a two-line count of how a kept digit vector can sum to s . Second, the carry map $c \mapsto (c + D - s)/3$ contracts: its fixed point is $D/2$, integrality is a wall, and every carry reachable from zero obeys $|c| \leq \lfloor (D - 1)/2 \rfloor$. Third, the design is invariant under $v \mapsto (2, \dots, 2) - v$, so $P_D[s] = P_D[2D - s]$, the carry reflection $c \mapsto -c$ commutes with the transfer matrix, and the orbit of the start state lives inside its $+1$ eigenspace, whose dimension is $\lfloor (D - 1)/2 \rfloor + 1 = \lceil D/2 \rceil$.

What is not new. That some constant-coefficient recurrence exists is free and standard. A finite carry state space makes $a_D(L)$ a 3-regular sequence in the sense of Allouche and Shallit [2], and the untrimmed automaton on states $-D, \dots, D$ already hands over order $2D + 1$ with no argument at all. A paper that led with “these slices obey a linear recurrence” would be selling a textbook fact. The object is not new either: higher-dimensional Menger continua are classical, and Hocking [6] cuts exactly the $D = 4$ diagonal for its geometry and its pictures. The $D = 3$ integers are catalogued as OEIS A299916 [11], whose Menger reading is a user comment counting the star-shaped holes rather than the tiles. The carry-automaton frame is classical too: with unrestricted digits it is the carries Markov chain of Holte’s “amazing matrix” [7], whose spectrum is known exactly; the design’s joint digit restriction and pinned output digit take the matrix here outside that theory, and the precise relationship is spelled out in [4].

What is new. The exact factorisation of P_D , the sharp order $\lceil D/2 \rceil$ proved for every $D \geq 2$, the exactness of that order verified through $D = 24$, the closed form of the reduced matrix, and every census at $D \geq 4$. Roughly a factor of four, uniformly in the dimension. Then, from the same factorisation read on the unit circle: an exact trigonometric product formula for a relaxed census (Theorem 3.16), and with it the odd half of the sign law $-\rho_D > f_D/3$ for every odd $D \not\equiv 1 \pmod{3}$, proved (Theorem 3.18), a statement that had resisted spectral routes because the relevant gap closes like $4/D$ (Conjecture 3.25).

The cold anchor. At $D = 3$ the construction, which never mentions a hexagon or a triangle, returns the 2×2 integer matrix $\begin{pmatrix} 6 & 6 \\ 1 & 3 \end{pmatrix}$, characteristic polynomial $\lambda^2 - 9\lambda + 12$, dominant root $(9 + \sqrt{33})/2$ and census 1, 6, 42, 306, 2250, 16578, 122202. Abel's tile substitution $\begin{pmatrix} 6 & 1 \\ 6 & 3 \end{pmatrix}$ [1] has the same trace 9 and determinant 12, A299916 has signature $(9, -12)$ and the same integers, and $\log_3((9 + \sqrt{33})/2) = 1.818410$ is Abel's published dimension. Two different routes to one spectrum is the reason to believe the machine in the dimensions where nobody has looked.

2. DEFINITIONS

Definition 2.1. The D -dimensional Menger design is

$$F_D = \{v \in \{0, 1, 2\}^D : \#\{i : v_i = 1\} \leq 1\},$$

the digit vectors with at most one middle coordinate. Its *fill* is $|F_D|$.

Definition 2.2. For $L \geq 0$ write each $x_i \in \{0, \dots, 3^L - 1\}$ in base 3 as $x_i = \sum_{j=0}^{L-1} v_i^{(j)} 3^j$. The *level- L Menger analog* is

$$S_L^{(D)} = \{x : v^{(j)} \in F_D \text{ for every } j\},$$

and the *central diagonal census* is

$$a_D(L) = \#\{x \in S_L^{(D)} : \sum_i x_i = D(3^L - 1)/2\}, \quad a_D(0) = 1.$$

The hyperplane is the one through the centre of the grid: the coordinate sum of the centre cell of $\{0, \dots, 3^L - 1\}^D$ is $D(3^L - 1)/2$. At $D = 3$ this is the diagonal cut of the classical sponge.

Definition 2.3. The *digit polynomial* is

$$P_D(t) = \sum_{v \in F_D} t^{v_1 + \dots + v_D} = \sum_{s=0}^{2D} B_D(s) t^s.$$

Definition 2.4. The *transfer matrix* $M^{(D)}$ acts on carry states $c \in \mathbb{Z}$ by

$$M_{c'c}^{(D)} = B_D(c + D - 3c'),$$

which is nonzero only when $0 \leq c + D - 3c' \leq 2D$. The *core* is $C_D = \{c \in \mathbb{Z} : |c| \leq r_D\}$ with $r_D = \lfloor (D - 1)/2 \rfloor$, and $M_{\text{even}}^{(D)}$ is the matrix of

$M^{(D)}$ restricted to the core and expressed in the reflection-symmetric basis $v_0 = e_0$, $v_k = e_k + e_{-k}$ for $1 \leq k \leq r_D$. Its size is $n_D = r_D + 1 = \lceil D/2 \rceil$, and χ_D denotes its characteristic polynomial.

3. RESULTS

Theorem 3.1. *For every $D \geq 2$ the sequence $a_D(L)$ satisfies the linear recurrence with constant integer coefficients given by χ_D , the characteristic polynomial of $M_{\text{even}}^{(D)}$. In particular a_D satisfies a constant-coefficient linear recurrence of order at most $\lceil D/2 \rceil$.*

Lemma 3.2. *For every $D \geq 1$, $P_D(t) = (1 + t^2)^{D-1}(1 + Dt + t^2)$, the fill is $|F_D| = 2^{D-1}(D + 2)$, and*

$$B_D(2k) = \binom{D}{k}, \quad B_D(2k+1) = D \binom{D-1}{k},$$

with the convention that a binomial coefficient with an out-of-range lower index is zero.

Lemma 3.3. *Every carry state reachable from $c = 0$ lies in the core C_D , and the core is forward invariant: if $|c| \leq r_D$ and $M_{c_c}^{(D)} \neq 0$ then $|c'| \leq r_D$.*

Lemma 3.4. *$P_D[s] = P_D[2D - s]$ for all s , the reflection $R : e_c \mapsto e_{-c}$ satisfies $RM^{(D)} = M^{(D)}R$, and $Re_0 = e_0$. Consequently the orbit $\{(M^{(D)})^L e_0 : L \geq 0\}$ lies in the $+1$ eigenspace of R inside the core, a space of dimension $\lceil D/2 \rceil$.*

Lemma 3.5. *Let $\omega = e^{2\pi i/3}$. Then $P_D(\omega) = (-1)^{D-1}\omega^D(D-1)$, so $|P_D(\omega)| = D-1$ for every $D \geq 1$.*

Lemma 3.6. *Write $\varepsilon_D = (D-1)(-1)^{D-1}/3$ and $f_D = 2^{D-1}(D+2)$. For $j \in \{0, 1, 2\}$,*

$$\sum_{s \equiv j \pmod{3}} B_D(s) = \begin{cases} f_D/3 + 2\varepsilon_D, & j \equiv D \pmod{3}, \\ f_D/3 - \varepsilon_D, & \text{otherwise.} \end{cases}$$

Proposition 3.7. *Fix an integer offset m with $|m| \leq r_D$ and let $a_D(L; m)$ count the cells of $S_L^{(D)}$ on the parallel hyperplane $\sum_i x_i = D(3^L - 1)/2 + m$. Then $a_D(\cdot; m)$ satisfies the same recurrence as a_D , the one given by χ_D , and $a_D(L; m) = a_D(L; -m)$ for every L .*

Remark 3.8. The hypothesis $|m| \leq r_D$ is sharp at even D . At $D = 4$ the offset $m = 2 = D/2$ sits exactly on the fixed point of the carry map, never enters the core, and the residual of χ_4 applied to $a_4(\cdot; 2)$ is the constant 128 rather than zero, for every window checked by `scripts/verify.py`.

Corollary 3.9. *$\limsup_L a_D(L)^{1/L}$ is the modulus of a root of χ_D , hence an algebraic number of degree at most $\lceil D/2 \rceil$. Whenever the counting exponent $\dim_{\text{slice}}(D) = \lim_L \log_3 a_D(L)/L$ exists, it is $\log_3$ of that algebraic number.*

Fact 3.10. For every D with $2 \leq D \leq 24$ the Hankel determinant $H_D = \det[a_D(i+j+1)]_{0 \leq i,j < n_D}$, computed in exact rational arithmetic, is nonzero; hence the minimal order of a_D is exactly $\lceil D/2 \rceil$ for those D . The first values are $H_2 = 2$, $H_3 = 72$, $H_4 = -6336$, $H_5 = -1029600000$, $H_6 = -62272025640000$, and H_{24} is a nonzero integer of 899 digits. Checked by `scripts/verify.py`.

Proposition 3.11. For every $D \geq 2$ the set of carry states reachable from $c = 0$ is exactly the core C_D , of size $D - 1$ at even D and D at odd D . Moreover M_C is irreducible and aperiodic: every state reaches 0 in one step, 0 loops to itself, and 0 reaches every state.

Fact 3.12. For every D with $2 \leq D \leq 24$,

$$\text{tr } M_{\text{even}}^{(D)} = \begin{cases} 3 \cdot 2^{D-2} - 1, & D \text{ even,} \\ 3D \cdot 2^{D-3}, & D \text{ odd.} \end{cases}$$

The even values are 2, 11, 47, 191, 767, 3071, 12287, ... and the odd values are 9, 60, 336, 1728, 8448, 39936, Checked by `scripts/verify.py`.

Fact 3.13. At $D = 3$, $M_{\text{even}}^{(3)} = \begin{pmatrix} 6 & 6 \\ 1 & 3 \end{pmatrix}$ with $\chi_3(\lambda) = \lambda^2 - 9\lambda + 12$, dominant root $(9 + \sqrt{33})/2$ and census 1, 6, 42, 306, 2250, 16578, 122202, matching OEIS [A299916](#) term for term and its signature (9, -12), and $\log_3((9 + \sqrt{33})/2) = 1.818410$ to six places. At $D = 2$ the census is 2^L , of order 1. At $D = 4$, $\chi_4(\lambda) = \lambda^2 - 11\lambda - 66$ with dominant root $(11 + \sqrt{385})/2$ and census 1, 6, 132, 1848, 29040, 441408, 6772128. At $D = 5$ and $D = 6$ the censuses begin 1, 30, 1000, 35700, 1321600 and 1, 20, 4030, 242300, 24642700. Checked by `scripts/verify.py`, which also confirms that three independent generators, brute-force digit enumeration, the substitution product $\prod_{j < L} P_D(t^{3^j})$, and the untrimmed carry automaton on states $-D, \dots, D$, agree term for term for $2 \leq D \leq 7$ and $0 \leq L \leq 4$.

Definition 3.14. The *diagonal sheaf census* is

$$b_D(L) = \# \left\{ x \in S_L^{(D)} : \sum_i x_i \equiv D(3^L - 1)/2 \pmod{3^L} \right\}, \quad b_D(0) = 1 :$$

the total census of the central hyperplane together with its translates by multiples of 3^L .

Lemma 3.15. For every real ψ ,

$$P_D(e^{i\psi}) = e^{iD\psi} \Phi_D(\psi), \quad \Phi_D(\psi) = (2 \cos \psi)^{D-1} (D + 2 \cos \psi),$$

and Φ_D is real-valued. For odd D , $\Phi_D(\psi) \geq 0$ for every ψ ; for even $D > 2$, $\Phi_D(\psi) < 0$ exactly when $\cos \psi < 0$.

Theorem 3.16. For every $D \geq 2$ and $L \geq 0$,

$$b_D(L) = \sum_{c \in C_D} (M_C^L)_{c0} = \frac{1}{3^L} \sum_{m=0}^{3^L-1} \prod_{j=0}^{L-1} \Phi_D\left(\frac{2\pi m 3^j}{3^L}\right).$$

In particular the right-hand side, a sum of 3^L products of cosines, is a positive integer.

Corollary 3.17. For $k \geq 1$ let $U_k = \{1 \leq u < 3^k : 3 \nmid u\}$ and

$$W_k = \sum_{u \in U_k} \prod_{j=0}^{k-1} \Phi_D\left(\frac{2\pi u 3^j}{3^k}\right).$$

Then $3^L b_D(L) = \sum_{k=0}^L f_D^{L-k} W_k$ with $W_0 = 1$, equivalently

$$3 b_D(k) - f_D b_D(k-1) = 3^{1-k} W_k \quad (k \geq 1).$$

Moreover the innermost factor of every product in W_k is $\Phi_D(2\pi u/3)$ with $3 \nmid u$, and

$$\Phi_D(2\pi/3) = \Phi_D(4\pi/3) = (-1)^{D-1}(D-1),$$

so $W_k = (-1)^{D-1}(D-1) V_k$ with $V_k = \sum_{u \in U_k} \prod_{j=0}^{k-2} \Phi_D(2\pi u 3^j/3^k)$.

Theorem 3.18. Let ρ_D be the spectral radius of $M_{\text{even}}^{(D)}$; it is a root of χ_D , and it is the dominant one. For every odd $D \geq 3$,

$$\rho_D \geq \frac{f_D}{3},$$

and if moreover $D \not\equiv 1 \pmod{3}$, the inequality is strict:

$$\rho_D > \frac{f_D}{3} \quad (D \text{ odd, } D \equiv 0, 2 \pmod{3}).$$

This is the odd half of Conjecture 3.22 for two of the three residue classes.

Fact 3.19. For the remaining residue class $D \equiv 1 \pmod{3}$, the integer $\det(f_D I - 3M_{\text{even}}^{(D)})$ is nonzero for every $D \leq 80$, checked in exact integer arithmetic by `scripts/verify.py`; with Theorem 3.18 this gives $\rho_D > f_D/3$ for every odd $D \leq 80$ with no exception. Two structural notes, both checked on the same sweep: for $D \equiv 1 \pmod{3}$ the 3-adic valuation of the determinant is at least n_D , so $\chi_D(f_D/3)$ is a nonzero integer there; and its valuation is otherwise erratic (2, 7, 6, 10, 17, 11, 11, 34, 16, 17, 33, 20, 20 over $D \equiv 1 \pmod{3}$, $4 \leq D \leq 40$, both parities), so no uniform 3-adic argument is in sight for this class.

Theorem 3.20 (the pinning). For every $D \geq 2$,

$$\left| \rho_D - \frac{f_D}{3} \right| \leq \frac{2(D-1)}{3},$$

refined by parity to $f_D/3 - 2(D-1)/3 \leq \rho_D \leq f_D/3 + (D-1)/3$ at even D and $f_D/3 - (D-1)/3 \leq \rho_D \leq f_D/3 + 2(D-1)/3$ at odd D . In particular $\rho_D = (f_D/3)(1 + O(D^2 2^{-D}))$, so

$$\log_3 \rho_D - (\log_3 f_D - 1) \longrightarrow 0 :$$

whatever the sign law does, the counting exponent of the recurrence converges to the generic Marstrand–Mattila value $\log_3 f_D - 1$, and the central diagonal

plane becomes asymptotically generic. Only the side of the approach remains in question, and only at even D .

Proposition 3.21 (the mass identity). *Let v be the Perron vector of M_C , unique up to scale by Proposition 3.11, and let $p_D = \sum_{c \in C_D, 3|c} v_c / \sum_{c \in C_D} v_c$ be its mass on carries divisible by 3. Then, exactly,*

$$\begin{aligned} 3\rho_D &= f_D + (-1)^{D-1}(D-1)(3p_D-1), \\ \text{i.e.} \quad \text{sgn}\left(\rho_D - \frac{f_D}{3}\right) &= (-1)^{D+1} \text{sgn}\left(p_D - \frac{1}{3}\right). \end{aligned}$$

Consequently the full sign law, both parities at once, is equivalent to the parity-free statement that $p_D > \frac{1}{3}$ for every $D \geq 2$: the Perron carry vector always puts more than a third of its mass on the carries divisible by three.

Conjecture 3.22. *For every even $D \geq 2$, and for every odd $D \equiv 1 \pmod{3}$ beyond the verified range,*

$$\text{sgn}(\rho_D - f_D/3) = (-1)^{D+1}, \quad f_D = 2^{D-1}(D+2).$$

The comparison value $\log_3 f_D - 1$ is the generic one: for a set of Hausdorff dimension s in $\mathbb{R}^D$, almost every hyperplane meets it in dimension at most $s-1$, with equality on a positive-measure set of translates once $s > 1$ [9, 10], and $\log_3 f_D$ is the dimension of the solid. The central diagonal hyperplane is one specific, highly symmetric slice, to which no such theorem applies; together with Theorem 3.18 the conjecture says its counting exponent misses the generic value in every dimension, alternately above and below. Evidence: the sign alternates without exception for $2 \leq D \leq 20$ computed from the roots of χ_D , and the exact rational sign of $\chi_D(f_D/3)$ alternates as $(-1)^D$ for $2 \leq D \leq 30$, both in `scripts/verify.py`. The sharpest reduced target on the even side is a two-step census contraction: for even D , exact computation finds $9b_D(2j+2) \leq f_D^2 b_D(2j)$ and $3b_D(2j+1) < f_D b_D(2j)$ with no violation through index 400 for every even $D \leq 50$, with the ratio $-W_{k+1}/(f_D W_k)$ at odd k peaking near $(D-2)/(D+2)$, a uniform margin. By induction these two inequalities give $b_D(L) \leq (f_D/3)^L$, which with Proposition 3.11 (irreducibility) and the nonvanishing of Fact 3.19 would give the even half; neither inequality is proved. Equivalently, by Proposition 3.21, everything left of the law – the even half and the residue class $D \equiv 1 \pmod{3}$ – is the single parity-free inequality $p_D > 1/3$. Failure mode, and why the even half is genuinely harder: for odd D the integrand Φ_D of Theorem 3.16 is non-negative, and Theorem 3.18 follows; for even D it is not, and the cancellation is real. Already $V_2 = 2[\Phi_D(2\pi/9) + \Phi_D(4\pi/9) + \Phi_D(8\pi/9)]$ is negative for every even $D \geq 6$, because $\Phi_D(8\pi/9) = -(2\cos(\pi/9))^{D-1}(D-2\cos(\pi/9))$ outweighs the positive terms, $2\cos(\pi/9) > 2\cos(2\pi/9)$. And no finite sign table of the W_k can decide the even half: since $b_D(L) \sim C\rho_D^L$ with $C > 0$, Corollary 3.17 forces $W_L \sim C3^{L-1}\rho_D^{L-1}(3\rho_D - f_D)$, so the eventual sign of W_k is the assertion $\rho_D < f_D/3$ itself; the signs observed at small k (they alternate transiently at base 3) are no evidence either way.

Remark 3.23 (note added, August 2026). The even half of Conjecture 3.22 has since been proved, in every even dimension at once, at base 3 and at base 5: see the companion paper [4], which builds exactly the second idea asked for above – a Collatz–Wielandt certificate closed analytically at depth $\Theta(D^2)$, the transient identified in closed form. The statement above is kept as written, both for the record and because its remaining content, strictness at odd $D \equiv 1 \pmod{3}$ beyond the verified range, is still open; [4] reduces it to a single 2-adic valuation bound.

Remark 3.24. The size of the excess appears to be governed by the same tower of angles. Numerically,

$$\rho_D - \frac{f_D}{3} = (-1)^{D+1} \frac{2(D-1)}{3} \prod_{k \geq 2} \left(\cos \frac{2\pi}{3^k} \right)^{D-1} \frac{D + 2 \cos(2\pi/3^k)}{D+2} \cdot (1+o(1)),$$

with the ratio of the two sides within 10^{-8} of 1 at $D = 61$, where the left side is computed by exact rational bisection of χ_{61} against the bracket $\chi_{61}(f_{61}/3) < 0 < \chi_{61}(f_{61}/3+1)$ in `scripts/verify.py`. The per-dimension decay constant is then $\prod_{k \geq 2} \cos(2\pi/3^k) = 0.7428747134\dots$, and the linear factor $D-1$ is the parity factor of Corollary 3.17. This asymptotic is a conjecture supported by computation; only the inequalities of Theorem 3.18 carry theorem dress.

Conjecture 3.25. *There is no uniform spectral gap: writing λ_2 for a second-largest eigenvalue of $M_{\text{even}}^{(D)}$ in modulus,*

$$\frac{\rho_D}{|\lambda_2|} = \frac{D+2}{D-2} + O(D^{-3}) \longrightarrow 1.$$

Evidence: the computed ratios fall from 3.551785 at $D = 4$ to 1.222222 at $D = 20$, monotonically from $D = 6$ on; over $6 \leq D \leq 20$ they agree with $(D+2)/(D-2)$ to within 0.05, and to six decimal places by $D = 18$. The two smallest dimensions in that range sit outside the agreement, 3.551785 against 3 at $D = 4$ and 2.253050 against 2.333333 at $D = 5$, which is where the $O(D^{-3})$ correction is largest. `scripts/verify.py` asserts the monotonicity, and the agreement over $6 \leq D \leq 20$. Failure mode: the ratios are floating-point roots of exactly computed integer polynomials, so a genuine near-degeneracy at some larger D could be misread; and the fitted shape $(D+2)/(D-2)$ is an observed match, not a derivation. This is the lighthouse that explains why Theorem 3.18 had to go around the spectrum. The classical route to the sign law reads it off $\text{sgn } \chi_D(f_D/3) = \prod_i \text{sgn}(f_D/3 - \lambda_i)$, which reports on ρ_D alone only if every subdominant eigenvalue sits below $f_D/3$; any argument pinning that has to work inside a margin that shrinks like $4/D$, and so cannot be a fixed-size perturbation estimate. Four natural routes were tried and refuted by their own computations: $M_{\text{even}}^{(D)}$ is not banded plus low rank, not totally nonnegative, not symmetrizable by a positive diagonal, and $f_D/3 \cdot I - M_{\text{even}}^{(D)}$ is not an M -matrix. The census route of Theorem 3.18 never

mentions λ_2 , which is how the odd half fell; the even half of Conjecture 3.22 still awaits either this separation or a second idea of the same kind.

4. PROOFS

Proof of Lemma 3.2. Split F_D by the number of middle coordinates. A vector with no coordinate equal to 1 chooses 0 or 2 in each of the D places, contributing $(1+t^2)^D$ to the generating function. A vector with exactly one coordinate equal to 1 chooses which coordinate, in D ways, and then 0 or 2 in the remaining $D-1$ places, contributing $Dt(1+t^2)^{D-1}$. No other vector lies in F_D , so

$$P_D(t) = (1+t^2)^D + Dt(1+t^2)^{D-1} = (1+t^2)^{D-1}(1+Dt+t^2).$$

Setting $t = 1$ gives $|F_D| = 2^{D-1}(D+2)$. For the coefficients, expand $(1+t^2)^{D-1}(1+t^2) = (1+t^2)^D$ for the even part and $Dt(1+t^2)^{D-1}$ for the odd part: the coefficient of t^{2k} is $\binom{D}{k}$ and that of t^{2k+1} is $D\binom{D-1}{k}$. $\square$

Proof of Theorem 3.1. Step 1: the census is a coefficient. By Definition 2.2 a cell of $S_L^{(D)}$ is exactly a choice of $v^{(0)}, \dots, v^{(L-1)} \in F_D$, and its coordinate sum is $\sum_i x_i = \sum_{j < L} 3^j \sigma_j$ with $\sigma_j = \sum_i v_i^{(j)}$. The number of vectors in F_D with digit sum σ is $B_D(\sigma)$, so for any target T ,

$$\#\{x \in S_L^{(D)} : \sum_i x_i = T\} = [t^T] \prod_{j=0}^{L-1} P_D(t^{3^j}).$$

Step 2: the carry recursion. For $c \in \mathbb{Z}$ let $N_L(c) = [t^{TL(c)}] \prod_{j < L} P_D(t^{3^j})$ with $T_L(c) = D(3^L - 1)/2 + c$, so $a_D(L) = N_L(0)$ and $N_0(c) = \mathbf{1}[c = 0]$. Peel the least significant digit vector, of digit sum s . Since

$$D(3^L - 1)/2 = 3 \cdot D(3^{L-1} - 1)/2 + D,$$

the remaining target after dividing by 3 is $T_{L-1}(c')$ with $c' = (c + D - s)/3$, and the peel is possible exactly when $3 \mid c + D - s$. Therefore

$$N_L(c) = \sum_{\substack{0 \leq s \leq 2D \\ s \equiv c+D \pmod{3}}} B_D(s) N_{L-1}\left(\frac{c+D-s}{3}\right),$$

which in the notation of Definition 2.4 reads $N_L = (M^{(D)})^L$ applied in the sense $a_D(L) = ((M^{(D)})^L)_{00}$: the entry $M_{c'c}^{(D)} = B_D(c+D-3c')$ is the weight of the step from source c to destination c' , and only finitely many entries are nonzero in each row and column because $0 \leq c + D - 3c' \leq 2D$.

Step 3: contraction. This is Lemma 3.3, proved below. It gives an invariant finite state set, the core C_D , containing the start state 0, so $a_D(L) = (M_C^L)_{00}$ where M_C is the finite matrix $M^{(D)}$ restricted to C_D .

Step 4: halving. This is Lemma 3.4, proved below. The vector $u_L = M_C^L e_0$ lies, for every L , in the $+1$ eigenspace E of the reflection R inside $\mathbb{R}^{C_D}$, and E is M_C -invariant because R and M_C commute. A basis of E is $v_0 = e_0$ and $v_k = e_k + e_{-k}$ for $1 \leq k \leq r_D$, so $\dim E = r_D + 1 = \lceil D/2 \rceil = n_D$. Let $M_{\text{even}}^{(D)}$

be the matrix of $M_C|_E$ in that basis and χ_D its characteristic polynomial, of degree n_D . By Cayley–Hamilton, $\chi_D(M_C|_E) = 0$, hence for every $L \geq 0$

$$\sum_{k=0}^{n_D} \chi_{D,k} u_{L+n_D-k} = \chi_D(M_C|_E) u_L = 0,$$

where $\chi_D(\lambda) = \sum_k \chi_{D,k} \lambda^{n_D-k}$ with $\chi_{D,0} = 1$. Applying the linear functional $w \mapsto w_0$ to this vector identity gives

$$\sum_{k=0}^{n_D} \chi_{D,k} a_D(L + n_D - k) = 0,$$

a constant-coefficient linear recurrence of order $n_D = \lceil D/2 \rceil$ satisfied by a_D . The coefficients are integers because $M_{\text{even}}^{(D)}$ has integer entries. $\square$

Proof of Lemma 3.3. If $M_{c'_c}^{(D)} \neq 0$ then $c' = (c + D - s)/3$ for some $0 \leq s \leq 2D$, so $c + D - s$ lies in $[c - D, c + D]$ and

$$|c'| \leq \frac{|c| + D}{3}.$$

Suppose $|c| \leq r_D = \lfloor (D-1)/2 \rfloor \leq (D-1)/2$. Then

$$|c'| \leq \frac{(D-1)/2 + D}{3} = \frac{3D-1}{6} = \frac{D}{2} - \frac{1}{6} < \frac{D}{2}.$$

Since c' is an integer, $|c'| < D/2$ forces $|c'| \leq \lfloor (D-1)/2 \rfloor = r_D$, whether D is even or odd: at even D an integer below $D/2$ is at most $D/2 - 1 = r_D$, and at odd D it is at most $(D-1)/2 = r_D$. So the core is forward invariant. The start state $c = 0$ lies in the core because $r_D \geq 0$ for $D \geq 2$, and induction on the number of steps finishes the claim. $\square$

Proof of Proposition 3.11. Containment in the core is Lemma 3.3. For the reverse, note first that every coefficient $B_D(s)$ with $0 \leq s \leq 2D$ is positive, by the closed forms of Lemma 3.2: $B_D(2k) = \binom{D}{k} > 0$ for $0 \leq k \leq D$ and $B_D(2k+1) = D \binom{D-1}{k} > 0$ for $0 \leq k \leq D-1$. Hence from carry c the one-step reachable set is exactly the set of integers in $[(c-D)/3, (c+D)/3]$, an interval that always contains 0 since $|c| \leq r_D < D$. If the states reachable in at most t steps include the interval $[-m, m]$, then one more step reaches every integer in $[-(m+D)/3, (m+D)/3]$, because the one-step intervals of consecutive sources overlap. So the reachable radius climbs as $m \mapsto \lfloor (m+D)/3 \rfloor$, starting from $\lfloor D/3 \rfloor$; while $m \leq r_D - 1 \leq (D-3)/2$ we have $m+D \geq 3m+3$, so each step gains at least 1, and by Lemma 3.3 the climb stops exactly at r_D . Irreducibility follows since every state reaches 0 in one step and 0 reaches every state; aperiodicity from the loop $M_{00}^{(D)} = B_D(D) > 0$. $\square$

Proof of Lemma 3.4. The involution $v \mapsto (2, \dots, 2) - v$ of $\{0, 1, 2\}^D$ preserves the number of coordinates equal to 1, so it permutes F_D and sends

digit sum s to $2D - s$. Hence $B_D(s) = B_D(2D - s)$ for all s . Now

$$\begin{aligned} M_{-c', -c}^{(D)} &= B_D(-c + D + 3c') = B_D(2D - (c + D - 3c')) \\ &= B_D(c + D - 3c') = M_{c', c}^{(D)}, \end{aligned}$$

which is exactly $RM^{(D)}R = M^{(D)}$, and $R^2 = I$ gives $RM^{(D)} = M^{(D)}R$. Clearly $Re_0 = e_0$. Therefore $R(M^{(D)})^L e_0 = (M^{(D)})^L Re_0 = (M^{(D)})^L e_0$ for every L , so the whole orbit lies in the $+1$ eigenspace of R . Restricted to the core, that eigenspace consists of the vectors with $w_c = w_{-c}$, spanned by e_0 and $e_k + e_{-k}$ for $1 \leq k \leq r_D$; these $r_D + 1 = \lceil D/2 \rceil$ vectors are visibly independent. $\square$

Proof of Lemma 3.5. Since $\omega^2 + \omega + 1 = 0$ we have $1 + \omega^2 = -\omega$ and $1 + D\omega + \omega^2 = (D - 1)\omega$. Substituting into Lemma 3.2,

$$P_D(\omega) = (-\omega)^{D-1}(D - 1)\omega = (-1)^{D-1}\omega^D(D - 1),$$

and $|\omega| = 1$ gives $|P_D(\omega)| = D - 1$. $\square$

Proof of Lemma 3.6. Roots of unity filter: for $j \in \{0, 1, 2\}$,

$$\sum_{s \equiv j} B_D(s) = \frac{1}{3} \sum_{m=0}^2 \omega^{-jm} P_D(\omega^m) = \frac{f_D}{3} + \frac{2}{3} \operatorname{Re}(\omega^{-j} P_D(\omega)),$$

using $P_D(1) = f_D$ and $P_D(\bar{\omega}) = \overline{P_D(\omega)}$ because P_D has real coefficients. By Lemma 3.5, $\omega^{-j} P_D(\omega) = (-1)^{D-1}(D - 1)\omega^{D-j}$, whose real part is $(-1)^{D-1}(D - 1)$ when $j \equiv D \pmod{3}$ and $-(-1)^{D-1}(D - 1)/2$ otherwise, since $\operatorname{Re} \omega^k$ is 1 for $k \equiv 0$ and $-1/2$ otherwise. Substituting $\varepsilon_D = (D - 1)(-1)^{D-1}/3$ gives $f_D/3 + 2\varepsilon_D$ in the distinguished class and $f_D/3 - \varepsilon_D$ in each of the other two. $\square$

Proof of Proposition 3.7. By Step 1 and Step 2 of the proof of Theorem 3.1 with target $T_L(m)$, the off-centre census is $a_D(L; m) = ((M^{(D)})^L)_{0m} = e_0^\top (M^{(D)})^L e_m$: the source state is the offset m and the accepted final state is 0. By Lemma 3.4, R commutes with $M^{(D)}$, is symmetric, and fixes e_0 , so

$$a_D(L; -m) = e_0^\top (M^{(D)})^L R e_m = (R e_0)^\top (M^{(D)})^L e_m = a_D(L; m).$$

Hence $a_D(L; m) = e_0^\top (M^{(D)})^L w$ with $w = \frac{1}{2}(e_m + e_{-m})$. Because $|m| \leq r_D$, the vector w lies in the $+1$ eigenspace E of R inside the core, which by Lemma 3.3 and Lemma 3.4 is invariant under M_C and has dimension n_D . So $(M_C)^L w \in E$ for all L , and Cayley–Hamilton applied to $M_C|_E$ annihilates the vector sequence exactly as in Step 4; taking the coordinate functional $w \mapsto w_0$ gives the same recurrence with the same coefficients. $\square$

Proof of Corollary 3.9. By Theorem 3.1, $a_D(L) = \sum_i p_i(L) \lambda_i^L$ where the λ_i are the roots of χ_D and the p_i are polynomials. Let Λ be the largest modulus among the roots that actually occur with a nonzero polynomial; it is positive, because every entry of $M^{(D)}$ is nonnegative and so $a_D(L) = ((M^{(D)})^L)_{00} \geq$

$(M_{00}^{(D)})^L = B_D(D)^L > 0$. Summing that expansion, the generating function $\sum_{L \geq 0} a_D(L)z^L$ is a rational function whose poles are exactly the points $1/\lambda_i$ over the occurring nonzero roots, so its radius of convergence is $1/\Lambda$, and Cauchy–Hadamard turns that into $\limsup_L a_D(L)^{1/L} = \Lambda$, the modulus of a root of χ_D . And χ_D is a monic integer polynomial of degree $n_D = \lceil D/2 \rceil$, so every root is an algebraic number of degree at most n_D . If the limit defining $\dim_{\text{slice}}(D)$ exists, then it equals $\log_3 \Lambda$ by taking logarithms. $\square$

Remark 4.1. Exactness of the order is a Hankel condition, not an eigenvalue condition. Given the upper bound n_D of Theorem 3.1, nonvanishing of $H_D = \det[a_D(i+j+1)]_{0 \leq i,j < n_D}$ forces the minimal order to be exactly n_D ; that is Fact 3.10. Indeed, if a_D obeyed some recurrence $a_D(L+m) = \sum_{k=1}^m c_k a_D(L+m-k)$ of order $m < n_D$ for every $L \geq 0$, then for every $i \geq m$ row i of that matrix would be the combination $\sum_{k=1}^m c_k \cdot (\text{row } i-k)$ of earlier rows, all n_D rows would lie in the span of the first m , and H_D would vanish. Only this direction holds: a vanishing H_D proves nothing, as the sequence $5, 2, 4, 8, 16, \dots$ shows, whose minimal order is 2 because $a(L+2) = 2a(L+1)$ for every $L \geq 0$ while no order-one recurrence fits, yet whose shifted Hankel determinant $\det\begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix}$ is zero. Distinct eigenvalues would not suffice either: writing $a_D(L) = \sum_i \lambda_i^L (e_0^\top \Pi_i e_0)$ with spectral projections Π_i , exactness also needs every weight $e_0^\top \Pi_i e_0$ to be nonzero. The two-line counterexample is $M = \text{diag}(1, 2)$ with start vector $e_0 = (1, 0)$: the eigenvalues are distinct and the matrix is non-derogatory, yet the sequence $e_0^\top M^L e_0 = 1$ has order 1, not 2. The Hankel determinant sees this where the eigenvalues do not, and it is what `scripts/verify.py` computes, in exact rational arithmetic.

Proof of Lemma 3.15. Write $1 + e^{2i\psi} = 2 \cos \psi e^{i\psi}$ and $1 + De^{i\psi} + e^{2i\psi} = e^{i\psi}(D + 2 \cos \psi)$. Substituting into the factorisation of Lemma 3.2,

$$P_D(e^{i\psi}) = (2 \cos \psi)^{D-1} e^{i(D-1)\psi} \cdot e^{i\psi}(D + 2 \cos \psi) = e^{iD\psi} \Phi_D(\psi).$$

For odd D the power $D-1$ is even, so $(2 \cos \psi)^{D-1} \geq 0$, and $D + 2 \cos \psi \geq D-2 > 0$ for $D \geq 3$. For even $D > 2$ the power is odd and $D + 2 \cos \psi > 0$, so the sign of Φ_D is the sign of $\cos \psi$. $\square$

Proof of Theorem 3.16. Step 1: the sheaf census is the free-end census. Set $N_p = D(3^p - 1)/2$, an integer for every D because $3^p - 1$ is even. The coordinate sums $T \equiv N_L \pmod{3^L}$ are exactly $T = N_L - 3^L c'$ over $c' \in \mathbb{Z}$, where terms with T outside $[0, D(3^L - 1)]$ vanish, so by Step 1 of the proof of Theorem 3.1,

$$b_D(L) = \sum_{c' \in \mathbb{Z}} [t^{N_L - 3^L c'}] \prod_{j=0}^{L-1} P_D(t^{3^j}).$$

Fix c' . A monomial choice from the product is a tuple $(s_0, \dots, s_{L-1})$ of digit sums with $\sum_j s_j 3^j = N_L - 3^L c'$, weighted by $\prod_j B_D(s_j)$. For $0 \leq p \leq L$ set $T_p = \sum_{j < p} s_j 3^j$ and $c_p = (N_p - T_p)/3^p \in \mathbb{Q}$; then $c_0 = 0$, the identity

$3c_{p+1} = c_p + D - s_p$ holds over $\mathbb{Q}$, and $c_p \in \mathbb{Z}$ exactly when $T_p \equiv N_p \pmod{3^p}$. The full congruence $T \equiv N_L \pmod{3^L}$ implies every partial one: $T_p \equiv T \equiv N_L \equiv N_p \pmod{3^p}$, since $N_L - N_p = D(3^L - 3^p)/2 = 3^p N_{L-p}$. So each tuple in the class determines an integer carry path from $c_0 = 0$ with final carry $c_L = c'$, stepping with the weights of Definition 2.4. Conversely, an integer carry path $0 = c_0 \rightarrow \dots \rightarrow c_L$ has $s_p = c_p + D - 3c_{p+1}$ and the same telescoping gives $T = N_L - 3^L c_L \equiv N_L \pmod{3^L}$. Hence the weighted count of tuples with final carry c' is $((M^{(D)})^L)_{c'0}$, paths out of the start state 0 (in Definition 2.4 the first index is the destination). By Lemma 3.3 no path from 0 leaves the core and M_C is a submatrix of $M^{(D)}$ with equal entries, so $b_D(L) = \sum_{c' \in C_D} (M_C^L)_{c'0} = \mathbf{1}^\top M_C^L e_0$.

Step 2: extraction. With $\zeta = e^{2\pi i/3^L}$ and $N_L = D(3^L - 1)/2$,

$$b_D(L) = \sum_{T \equiv N_L(3^L)} [t^T] \prod_{j < L} P_D(t^{3^j}) = \frac{1}{3^L} \sum_{m=0}^{3^L-1} \zeta^{-N_L m} \prod_{j=0}^{L-1} P_D(\zeta^{m3^j}).$$

Step 3: the phases cancel. Write $\psi_j = 2\pi m3^j/3^L$. By Lemma 3.15, $P_D(\zeta^{m3^j}) = e^{iD\psi_j} \Phi_D(\psi_j)$, and

$$\prod_{j=0}^{L-1} e^{iD2\pi m3^j/3^L} = \exp\left(2\pi i m D \frac{3^L-1}{2 \cdot 3^L}\right) = \zeta^{mN_L},$$

which cancels the extraction phase $\zeta^{-N_L m}$ exactly, leaving the stated real sum. It is a positive integer because it equals the census $b_D(L)$, and $b_D(L) \geq a_D(L) \geq B_D(D)^L \geq 1$. $\square$

Proof of Corollary 3.17. Sort the summation index m of Theorem 3.16 by its 3-adic valuation: $m = 0$ contributes $\Phi_D(0)^L = f_D^L$, and $m = 3^{L-k}u$ with $u \in U_k$, $1 \leq k \leq L$, has angles $2\pi u3^j/3^k$ reduced mod 2π : for $j \geq k$ the angle is an integer multiple of 2π and the factor is $\Phi_D(0) = f_D$; the remaining factors are $\prod_{j < k} \Phi_D(2\pi u3^j/3^k)$. Hence $3^L b_D(L) = \sum_{k=0}^L f_D^{L-k} W_k$, and subtracting f_D times the same identity at $L-1$ leaves $3^L b_D(L) - f_D 3^{L-1} b_D(L-1) = W_L$. For the innermost factor, $u3^{k-1}/3^k = u/3$ with $3 \nmid u$, and $2 \cos(2\pi/3) = 2 \cos(4\pi/3) = -1$ gives $\Phi_D = (-1)^{D-1}(D-1)$ there. $\square$

Proof of Theorem 3.18. The spectral radius is the dominant root. $M_{\text{even}}^{(D)}$ is a square matrix, so its spectral radius is the maximal modulus of the roots of χ_D ; that it is itself a root is Perron–Frobenius for nonnegative matrices (no irreducibility required): a nonnegative square matrix has its spectral radius as an eigenvalue, with an entrywise nonnegative eigenvector [8, Theorem 8.3.1]. Nonnegativity of $M_{\text{even}}^{(D)}$ holds because its entries are $B_D(D-3i)$ and $B_D(D+j-3i) + B_D(D-j-3i)$.

Step 1: the census grows at least like $(f_D/3)^L$. For odd D , $\Phi_D \geq 0$ everywhere by Lemma 3.15, so every $W_k \geq 0$, and Corollary 3.17 gives $3^L b_D(L) = \sum_k f_D^{L-k} W_k \geq f_D^L$, i.e. $b_D(L) \geq (f_D/3)^L$ for every L .

Step 2: the growth is bounded by the spectral radius of the core. $b_D(L)$ is the 0-column sum of M_C^L , so $b_D(L) \leq \|M_C^L\|_1$, the maximal column sum. By Gelfand's formula $\|M_C^L\|_1^{1/L} \rightarrow \rho(M_C)$, so $\rho(M_C) \geq \limsup_L b_D(L)^{1/L} \geq f_D/3$.

Step 3: the core radius is ρ_D . By Perron–Frobenius again there is $v \geq 0$, $v \neq 0$, with $M_C v = \rho(M_C) v$. Its even part $w = (v + Rv)/2$ is nonnegative, and nonzero: $w = 0$ would force $Rv = -v$ with both sides nonnegative, hence $v = 0$. Since R commutes with M_C (Lemma 3.4), w is an eigenvector of M_C at $\rho(M_C)$ lying in the $+1$ eigenspace E , so $\rho(M_C)$ is an eigenvalue of $M_{\text{even}}^{(D)} = M_C|_E$ and $\rho_D \geq \rho(M_C)$. Conversely every eigenvalue of $M_C|_E$ is an eigenvalue of M_C , so $\rho_D \leq \rho(M_C)$. Hence $\rho_D = \rho(M_C) \geq f_D/3$.

Step 4: strictness off the residue class $D \equiv 1$. The matrix $J = f_D I - 3M_{\text{even}}^{(D)}$ has integer entries, and

$$\chi_D(f_D/3) = \det((f_D/3)I - M_{\text{even}}^{(D)}) = 3^{-n_D} \det J.$$

Mod 3 the term $3M_{\text{even}}^{(D)}$ vanishes, so $\det J \equiv f_D^{n_D} \pmod{3}$. Since $f_D = 2^{D-1}(D+2)$, $3 \mid f_D$ exactly when $D \equiv 1 \pmod{3}$; for $D \not\equiv 1 \pmod{3}$ we get $\det J \not\equiv 0 \pmod{3}$, so $\chi_D(f_D/3) \neq 0$ and $f_D/3$ is not an eigenvalue. Since $\rho_D \geq f_D/3$ and ρ_D is an eigenvalue, $\rho_D > f_D/3$. $\square$

Proof of Theorem 3.20. By Lemma 3.3 the core is forward invariant, so no column of M_C loses mass under the restriction: the column of M_C at source c sums to $\sum_{s \equiv c+D \pmod{3}} B_D(s)$, which by Lemma 3.6 equals $f_D/3 + 2\varepsilon_D$ when $c \equiv 0 \pmod{3}$ and $f_D/3 - \varepsilon_D$ otherwise, with $\varepsilon_D = (D-1)(-1)^{D-1}/3$. The spectral radius of a nonnegative matrix lies between its smallest and largest column sums [8, Theorem 8.1.22], and $\rho(M_C) = \rho_D$ by Step 3 of the proof of Theorem 3.18. Reading off the two values by the parity of ε_D gives the refined brackets, and both are contained in $|\rho_D - f_D/3| \leq 2(D-1)/3$. Finally $2(D-1)/3$ against $f_D/3 = 2^{D-1}(D+2)/3$ gives the relative error $O(D^2 2^{-D})$, and $\log_3$ of $1 + O(D^2 2^{-D})$ tends to 0. $\square$

Proof of Proposition 3.21. By Proposition 3.11, M_C is irreducible, so its Perron vector $v \geq 0$ is unique up to scale and $M_C v = \rho_D v$ with $\rho(M_C) = \rho_D$ as above. Multiply by the all-ones row vector: $\rho_D \sum_c v_c = \mathbf{1}^\top M_C v = \sum_c \sigma(c) v_c$, where $\sigma(c)$ is the column sum at c , two-valued as in the proof of Theorem 3.20. Writing $\sigma(c) = f_D/3 + \varepsilon_D (3 \cdot \mathbf{1}[3 \mid c] - 1)$ and dividing by $\sum_c v_c$,

$$\rho_D = \frac{f_D}{3} + \varepsilon_D (3p_D - 1),$$

which is the identity. Multiplying $\rho_D - f_D/3 = (-1)^{D-1}(D-1)(p_D - \frac{1}{3})$ by $(-1)^{D+1}$ gives the sign statement, and the equivalence with the sign law follows because $(-1)^{D+1} \operatorname{sgn}(\rho_D - f_D/3) = \operatorname{sgn}(p_D - \frac{1}{3})$ at every D . $\square$

5. REPRODUCIBILITY

Two scripts, plain Python 3, no dependencies, no arguments, every path relative to the lane; run them from the lane root.

`scripts/verify.py` runs in about four seconds and dies loudly on the first mismatch, naming what it got and what it wanted. It covers, in order: the factorisation of Lemma 3.2 and its closed entry form against brute-force enumeration of all 3^D digit tuples for $2 \leq D \leq 8$; agreement of three independent generators, brute enumeration, the substitution product $\prod_{j < L} P_D(t^{3^j})$, and the untrimmed carry automaton on states $-D, \dots, D$, for $2 \leq D \leq 7$ and $0 \leq L \leq 4$; the exact reachable carry set by breadth-first search for $2 \leq D \leq 24$, corroborating Proposition 3.11; nonvanishing of the exact rational Hankel determinant for $2 \leq D \leq 24$, which is Fact 3.10 and Remark 4.1; the two trace formulas of Fact 3.12 for $2 \leq D \leq 24$; the fill and the three residue-class sums of Lemma 3.6 for $2 \leq D \leq 12$; Proposition 3.7 for $3 \leq D \leq 7$ at every offset $1 \leq m \leq r_D$ over windows through $L = 2n_D + 3$, together with the sharpness witness of Remark 3.8 at $D = 4$, $m = 2$; the anchors of Fact 3.13 at $D = 2, 3, 4, 5, 6$, including the matrix, its characteristic polynomial, the census, and $\log_3((9 + \sqrt{33})/2)$ to six places; the spectral ratios of Conjecture 3.25 for $4 \leq D \leq 20$, whose monotonicity and agreement with $(D+2)/(D-2)$ are asserted over $6 \leq D \leq 20$; and both sign checks of the sign law, floating-point roots for $2 \leq D \leq 20$ and the exact rational sign of $\chi_D(f_D/3)$ for $2 \leq D \leq 30$, now corroboration of Theorem 3.18 on the odd side and evidence for Conjecture 3.22 on the even side. The new results are covered by five further blocks: the sheaf census of Definition 3.14 computed three independent ways (free-end transfer count, coefficient-class sums of the substitution product in exact integers, and the trigonometric sum of Theorem 3.16 in floating point) for $2 \leq D \leq 7$; the step identity and parity factor of Corollary 3.17 in exact integers; strict positivity of every W_k , $k \leq 8$, at every odd $D \leq 25$, the mechanism of Theorem 3.18; nonvanishing of the exact integer $\det(f_D I - 3M_{\text{even}}^{(D)})$ for $2 \leq D \leq 80$ together with its congruence to $f_D^{n_D} \bmod 3$, which is Fact 3.19; and the sign of W_2 at even $6 \leq D \leq 24$, the obstruction quoted in Conjecture 3.22; and the mass identity of Proposition 3.21 together with both parity-refined brackets of Theorem 3.20, by power iteration on the core matrix for $2 \leq D \leq 20$. Characteristic polynomials are computed by the Faddeev–LeVerrier recursion over exact rationals; only the spectral checks, the product-formula cross-check and the $D = 3$ dimension use floating point, and the spectral checks do so on integer polynomials rescaled by $f_D/3$ so that all roots have modulus near one.

`scripts/figure.py` runs instantly and redraws `figures/order.svg`, the README’s copy of Fig. 1, from the same two formulas $2D + 1$ and $\lceil D/2 \rceil$.

The $D = 3$ census is OEIS A299916 [11]. The ladders at $D = 4, 5, 6$ are not in the OEIS to the author’s knowledge; no novelty search was run, and they are offered here only as computed values.

ACKNOWLEDGMENTS

This paper was developed and verified in collaboration with Claude (Anthropic). The author takes sole responsibility for every claim.

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